



CS343: Operating Systems
Quiz-2, November 7, 2024
Indian Institute of Technology Guwahati

Name: _____ **Roll Number:** _____

Answer all the questions with proper explanation

Q1. Consider the following snapshot of a system

	<u>Allocation</u>	<u>Max</u>
	<u>A B C D</u>	<u>A B C D</u>
T_0	3 0 1 4	5 1 1 7
T_1	2 2 1 0	3 2 1 1
T_2	3 1 2 1	3 3 2 1
T_3	0 5 1 0	4 6 1 2
T_4	4 2 1 2	6 3 2 5

Using the banker's algorithm, determine whether or not each of the following states is unsafe. If the state is safe, illustrate the order in which the threads may complete. Otherwise, illustrate why the state is unsafe. (2 + 2 = 4 marks)

a. Available = (0, 3, 0, 1)

b. Available = (1, 0, 0, 2)

Answer:

a. Not safe. Threads T_2 , T_1 , and T_3 are able to finish, but no remaining threads can finish.

b. Safe. Threads T_1 , T_2 , and T_3 are able to finish. Following this, threads T_0 and T_4 are also able to finish.

Q2. A 6000 rpm disk requests come to a disk driver for cylinders in the order 10, 22, 20, 2, 40, 6 and 38 and at a time when the disk drive is reading from cylinder 20. The seek time per cylinder is half the average rotational delay. Show the head movement and find the total seek time, if the disk arm scheduling algorithms is first-come-first-served. (3 marks)

Answer:

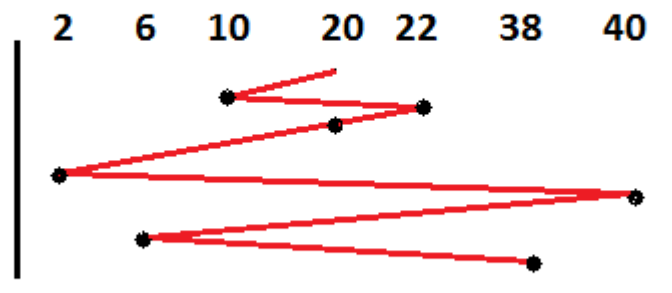
Time taken for one full rotation = $(60/6000) = 0.01s = 10 \text{ ms}$

Average rotational delay = $\frac{1}{2} \times \text{time taken for full rotation} = \frac{1}{2} \times 10\text{ms} = 5 \text{ ms}$

Seek time = 2.5 ms/ cylinder

In FCFS, the request which comes first to disk driver is served first.

Given cylinder Queue: 10, 22, 20, 2, 40, 6, 38



Head Movements: $(20 - 10) + (22 - 10) + (22 - 2) + (40 - 2) + (40 - 6) + (38 - 6) = 146$

For 146 seeks, seek time = $2.5 \times 146 = 365$ ms

Q3. Consider the following memory table consisting of holes and active processes:

P1	H1	P2	P3	H2	P4	H3	P5	H4
2K	9K	3K	6K	5K	21K	10K	11K	3K

- A new process P6 with size 5K is arriving, which hole will be assigned for P6, clearly depict by showing the memory table?
- Find out new holes after the termination of P2 and P5 (active process are P1, P3, P4, and P6), show changes in the memory table.

Solve the above two questions using (a) First, (b) Best and (c) Worst Fit memory allocation strategy?

(2 x 3 = 6 marks)

Answer:

a. First Fit

- P6 with size 5K is arriving: It will be allocated at H1, hence leaving H1 with 4K

P1	P6	H1	P2	P3	H2	P4	H3	P5	H4
2K	5K	4K	3K	6K	5K	21K	10K	11K	3K

- P2 and P5 Terminated

P1	P6	H1	P3	H2	P4	H3
2K	5K	7K	6K	5K	21K	24K

b. Best Fit

- P6 with size 5K is arriving: It will be allocated at H2

P1	H1	P2	P3	P6	P4	H2	P5	H3
2K	9K	3K	6K	5K	21K	10K	11K	3K

- P2 and P5 Terminated

P1	H1	P3	P6	P4	H2
2K	12K	6K	5K	21K	24K

c. **Worst Fit**

- I. P6 with size 5K is arriving: It will be allocated at the largest hole H3, hence leaving H3 with 5K

P1	H1	P2	P3	H2	P4	P6	H3	P5	H4
2K	9K	3K	6K	5K	21K	5K	5K	11K	3K

- II. P2 and P5 Terminated

P1	H1	P3	H2	P4	P6	H3
2K	12K	6K	5K	21K	5K	19K

Q4. There are 64-page frames in the physical memory of a system. It uses a paging scheme that employs 4KB pages. The page table for a program is indexed based on virtual page number and each page table entry contains the frame number and 2 control bits. The total size of page table is 512 bytes. Assume that the memory is byte addressable memory. (1 X 3 = 3 marks)

- How many bits are there in the physical address?
- What is the size of the page table entries in bits?
- How many bits are there in the virtual address?

Answer:

a) Physical memory = # frames x page size = $64 \times 4\text{KB} = 2^6 \times 2^{12} = 2^{18} \text{ B}$
Hence, there are 18 bits in physical address

b) Page table entries = Number of frame bits + control bits = $6 + 2 = 8$ bits

c) #bits in virtual address = Page number (p) + offset (d)

Offset is the page size, i.e 12, page number =?

Page number = Page table size / size of one page table entries
= $512 \text{ B} / 8 \text{ B} = 2^9$

Hence, there are 21 bits (9+12) in the virtual address

Q5. If a 7500 rpm disk drive has 256 sectors per track and the track seek time is $[(1/10) \times \text{average rotational delay} + 6.25 \text{ micro-second}]$, what is the cylinder skew? (2 marks)

Answer:

Time taken for one full rotation = $(60/7500) = 0.008 \text{ s} = 8 \text{ ms}$

Average rotational delay = $\frac{1}{2} \times \text{time taken for full rotation} = \frac{1}{2} \times 8 \text{ ms} = 4 \text{ ms}$

Track has 256 sectors

New sector in every $(8/256) = 31.25 \text{ microsecond}$

Track seek time is = $1/10 \times 4 \text{ ms} + 6.25 \text{ microsecond} = 400 + 6.25 = 406.25 \text{ microseconds}$

Cylinder skew = seek time/time taken for each sector = $406.25 / 31.25 = \mathbf{13 \text{ sectors}}$

Q6. Consider a file currently consisting of 50 blocks. Assume that the file control block is already in memory. Calculate how many disk I/O operations, i.e. number of reads and writes are required for contiguous allocation strategy, if, for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory. (2 marks)

- a. The block is added at the beginning
- b. After performing the above now we remove a block from position number 26.

Answer:

- a. Block 50 read and written to 51
Block 49 read and written to 50
...
Block 1 read and written to 2
New Block written to 1
Thus, we have 50 reads and 51 writes
- b. Total 51 Block
Block # 26 is removed
Block 27 is read and written to 26
Block 28 is read and written to 25
....
Block 51 is read and written to 50
Thus, we have 25 reads and 25 writes